

VIISNU ANAND S

ELECTRONICS AND COMMUNICATION ENGINEER

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SUMMARY

I am an Electronics and Communication Engineering Graduate whose passion is to talk about technology and innovation. And familiar with embedded systems, obtained both academically and professionally. I am an active and fast learner who is motivated by problem-solving and is willing to develop all the time and make a mark on the world of engineering and social development.

EDUCATION

Kumaraguru College of Technology B.E Electronics and Communication Engineering	2022–2026
Sri Chaitanya Techno School, Salem Higher Secondary Education (HSC)	2020–2022
Sri Chaitanya Techno School, Salem Secondary School Education (SSLC)	2018–2020

INTERNSHIP

Titan Engineering and Automation Limited (TEAL)	Mar 2026
• Gained practical experience in both SCADA systems and electrical assembly operations within an industrial manufacturing environment. • Supported process monitoring, system diagnostics, and industrial automation activities. • Assisted in electrical wiring, assembly, testing, and quality verification of industrial products. • Worked closely with cross-functional engineering teams to improve operational efficiency and maintain production standards. • Developed hands-on knowledge of industrial automation, electrical systems, manufacturing practices, and quality control procedures.	
Pricol Limited (Corporate Manufacturing Engineering (CME) Department)	June 2025
• Worked on developing a low-cost wire colour sequence detection system using Jetson Orin Nano, Arducam, and OpenCV. Gained hands-on exposure industrial R&D workflows, contributing to automation solutions in automotive manufacturing.	

PROJECTS

Smart India Hackathon 2024 – Winner of the Problem Statement (ISRO)

- Designed a DVB-S2X signal detection and classification AI-based real-time satellite communication analysis.
- Applied deep learning-based modulation classification, which is applied to SDR signal pipelines and used to perform proper identification and classification of DVB-S2X waveforms.
- Delivered high-performance in real-time that could be used in aerospace and satellite communications.
- Technologies: Python, Deep Learning, OpenCV, SDR Processing.

IRoC-U 2025 (ISRO Robotics Challenge)

- Cleared the Qualification Round and was among the 37 highest ranked teams in India.
- Came up with AI based autonomous UAV that can navigate in GNSS denied Mars-like environments.
- Introduced SLAM, LiDAR based mapping, 3D vision and sensor fusion to have precision navigation and autonomous landing.
- Onboard computing using integrated perception and flight control in real-time decision-making.
- Technologies: ROS, ZED 2i, Jetson Orin Nano, Pixhawk, OpenCV, LiDAR, Python.

Solar Power Generation and Storage Optimization (Final Year Project)

- Developed a data-driven framework for optimizing solar power generation and battery storage capacity in off-grid residential energy systems.
- Utilized NASA POWER API data and measured load profiles to model solar irradiance, energy demand, and storage requirements.
- Designed mathematical models for PV generation, battery sizing, and system reliability to improve energy utilization and reduce overall system cost.
- Developed a working prototype integrating solar panels, battery storage, and inverter systems for real-time energy management.
- Technologies: Python, NASA POWER API, Solar PV Systems, Battery Storage Systems, Inverters, Energy Modelling.

Wire Colour Detection System (Internship Project)

- Designed and developed a vision-based system of wire colour sequence detection of automotive wire harnesses.
- Helped to reduce overall system cost by adopting the Keyence camera, replacing higher-cost industrial vision solutions while maintaining reliable detection accuracy.
- Real-time industrial applications of deployed optimized OpenCV pipelines with embedded hardware.
- Technologies: Jetson Orin Nano, OpenCV, Python.

Gesture-Controlled Robot – Mini Project

- Developed a live-streaming surveillance robot to be used in remote monitoring and security.
- Real time video streaming with Pi camera and web control.
- Created Flask web interface to control movements and monitor them; motor control by means of L298N driver.
- Technologies: Raspberry Pi 4, OpenCV, Flask, HTML/CSS, L298N Motor Driver.

Surveillance Pi Bot – Mini Project

- Developed a live-streaming surveillance robot to be used in remote monitoring and security.
- Real time video streaming with Pi camera and web control.
- Created Flask web interface to control movements and monitor them.
- Technologies: Raspberry Pi 4, OpenCV, Flask, HTML/CSS, L298N Motor Driver.

SOFT SKILLS

• Time Management • Leadership	• Teamwork • Problem Solving
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TECHNICAL SKILLS

• C Programming • Embedded Systems	• NVIDIA Boards • GNU Radio
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LANGUAGES KNOWN

• Tamil • English	• Hindi – Beginner Level • French – Beginner Level
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CERTIFICATIONS AND ACHIEVEMENTS

Awards / Activities:

- Award for Extracurricular Excellence – Honoured by KCT for contributions beyond academics (Aug 2015).
- Award for Extracurricular Excellence – Honoured by KCT for contributions beyond academics (Jan 2026).

Certifications:

- NPTEL Certification – Introduction to IoT
- Emerging Trends in IoT Cloud Computing (Workshop)
- VLSI Essential Concepts and Detailed Interview Guide – Udemy